FY-924

Reg. No. :

Name : ...

FIRST YEAR HIGHER SECONDARY
IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

Time : 2 Hours

PHYSICS

Cool-off time : 15 Minutes

Maximum : 60 Scores

General Instructions to Candidates :

- There is a 'Cool-off time' of 15 minutes in addition to the writing time.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

വിദ്യാർത്ഥികൾക്കുള്ള പൊതുനിർദ്ദേശങ്ങൾ :

- നിർദ്ദിഷ്ട സമയത്തിന് പുറമെ 15 മിനിറ്റ് 'കൂൾ ഓഫ് ടൈം' ഉണ്ടായിരിക്കും.
- ചോദ്യങ്ങൾ പരിചയപ്പെടാനും ഉത്തരങ്ങൾ ആസൂത്രണം ചെയ്യാനും 'കൂൾ ഓഫ് ടൈം' ഉപയോഗിക്കുക.
- ഉത്തരങ്ങൾ എഴുതുന്നതിനു മുമ്പ് ചോദ്യങ്ങൾ ശ്രദ്ധാപൂർവം വായിക്കണം.
- നിർദ്ദേശങ്ങൾ മുഴുവനും ശ്രദ്ധാപൂർവം വായിക്കണം.
- കണക്ക് കൂട്ടലുകൾ, ചിത്രങ്ങൾ, ഗ്രാഫുകൾ, എന്നിവ ഉത്തരപേപ്പറിൽ തന്നെ ഉണ്ടായിരിക്കണം.
- ചോദ്യങ്ങൾ മലയാളത്തിലും നല്കിയിട്ടുണ്ട്.
- ആവശ്യമുള്ള സ്ഥലത്ത് സമവാക്യങ്ങൾ കൊടുക്കണം.
- പ്രോഗ്രാമുകൾ ചെയ്യാനാകാത്ത കാൽക്കുലേറ്ററുകൾ ഒഴികെയുള്ള ഒരു ഇലക്ട്രോണിക് ഉപകരണവും പരീക്ഷാഹാളിൽ ഉപയോഗിക്കുവാൻ പാടില്ല.

SECTION – A

Answer any 5 questions from 1 to 7. Each question carries 1 score. **(5 × 1 = 5)**

1. Which of the following has least significant figures ?

(a) 0.80760 (b) 0.80200

(c) 0.08076 ✓ (d) 80.200

2. State whether the given statement is true or false : **Answer : True**

"The sign of acceleration alone does not tell about the speed of the particle, either it is increasing or decreasing."

3. If $|A + B| = |A - B|$, the angle between the vectors A and B are **90^0** .

4. The average kinetic energy of gas molecule is proportional to

(a) Potential energy (b) Temperature ✓

(c) Density (d) Size

5. Stress required to double the length of a wire of Young's modulus Y is

(a) Y/2 (b) 2Y

(c) Y ✓ (d) 4Y

6. Viscosity of gases **Increase** with temperature, whereas viscosity of liquids **Decrease** with temperature. (Increase/Decrease).

7. A particle executing Simple Harmonic Motion is an example of

(a) Acceleration of constant magnitude and direction

(b) Acceleration of changing magnitude and direction ✓

(c) Acceleration of constant magnitude but changing direction

(d) Acceleration of changing magnitude but constant direction

SECTION – B

Answer any 5 questions from 8 to 14. Each question carries 2 scores. **(5 × 2 = 10)**

8. Can a body have zero velocity, still accelerating ? Explain.

(8) Yes. At the highest point of a ball thrown vertically upwards, its velocity becomes zero momentarily, but its acceleration due to gravity acts downwards. Hence, a body can have zero velocity and still have acceleration.

9. What is Poisson's ratio ? Explain its significance.

(9) Poisson's ratio is the ratio of lateral strain to longitudinal strain, with a negative sign.
It indicates how much a material contracts or expands sideways when stretched or compressed.

10. (a) Hydrostatic pressure is a _____ quantity. (Vector/Scalar) (1)

(b) Explain why ? (1)

(10) (a) Scalar

(b) Pressure has magnitude but no specific direction. At a point in a fluid at rest, pressure acts equally in all directions.

11. Triple point of carbon dioxide is –56.6 °C. Express this temperature on Fahrenheit scale.

(11)

$$F = \frac{9}{5} C + 32$$

Here, $C = -56.6\ ^\circ C$

Substitute in the formula,

$$F = \frac{9}{5}(-56.6) + 32$$

$$= -101.88 + 32$$

$$= -69.88\ ^\circ F$$

$$\therefore \quad -56.6\ ^\circ C = \boxed{-69.88\ ^\circ F}$$

12. State Kepler's law of areas. Write a mathematical expression of it.

(12) Kepler's second law states that the line joining a planet and the Sun sweeps out equal areas in equal intervals of time.

- **Mathematical expression: dA/dt = constant**

13. A body which is initially at rest undergoes one-dimensional motion. Show that the power delivered to it at a time 't' seconds is proportional to time 't'.

(13) Let the body (mass m) start from rest and move with constant acceleration a.

Then, velocity at time t is $v = u + at$

$= 0 + at$

$= at$

Force acting on the body, $F = ma$ (constant)

Instantaneous power delivered to the body,

$P = Fv = ma(at)$

$= ma^2t$

Since m and a are constants,

$P \propto t$

14. Distinguish between transverse and longitudinal waves.

(14)

TRANSVERSE WAVES	LONGITUDINAL WAVES
Particles vibrate perpendicular to the direction of wave propagation.	Particles vibrate parallel to the direction of wave propagation.
Crests and troughs are formed.	Compressions and rarefactions are formed.
Eg: Waves on a stretched string.	Example: Sound waves in air.

SECTION – C

Answer any 6 questions from 15 to 21. Each question carries 3 scores. **(6 × 3 = 18)**

15. (a) Velocity and Electric current cannot be added together. Name the principle behind this. **(1)**

(b) A boy recalls the relativistic variation of mass wrongly as $m = m_0 / \sqrt{(1 - v^2)}$, where '$m_0$' is the rest mass and 'v' is the velocity.

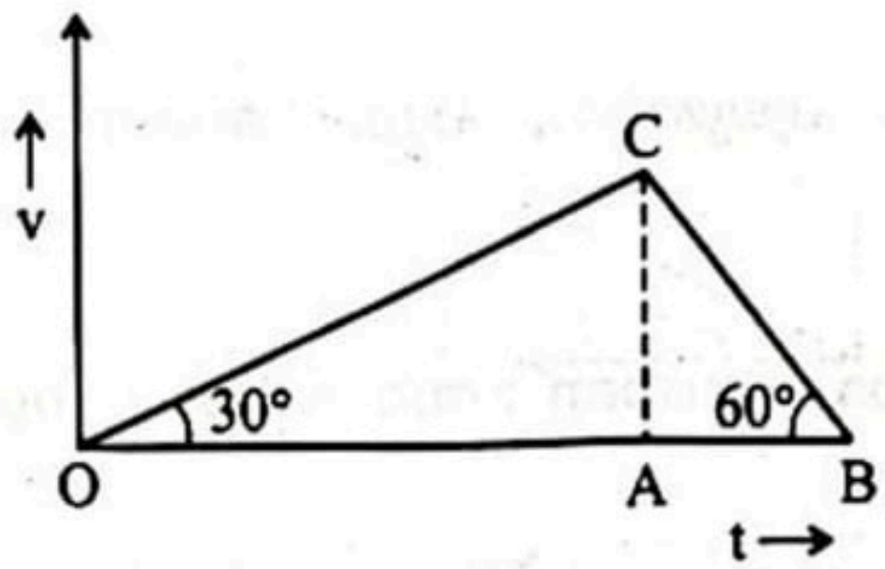
Using dimensional method, put the missing term velocity of light 'c' in proper place and correct the equation. **(2)**

(a) The principle of dimensional homogeneity.

(b) The given equation is missing the velocity of light .

The correct equation is: $$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

16. Study the velocity-time graph of a body given below and answer the following questions :



(a) Obtain the ratio of average acceleration during the intervals 'OA' and 'AB'. **(2)**

(b) The area of triangle OBC will represent which physical quantity ? **(1)**

(a) Average acceleration = slope of v–t graph.

In OA : $a_{OA} = \tan 30° = \frac{1}{\sqrt{3}}$

In AB : $a_{AB} = -\tan 60° = -\sqrt{3}$

(a)

Ratio : $a_{OA} : a_{AB} = \frac{1}{\sqrt{3}} : (-\sqrt{3}) = 1 : 3$

(b) Displacement

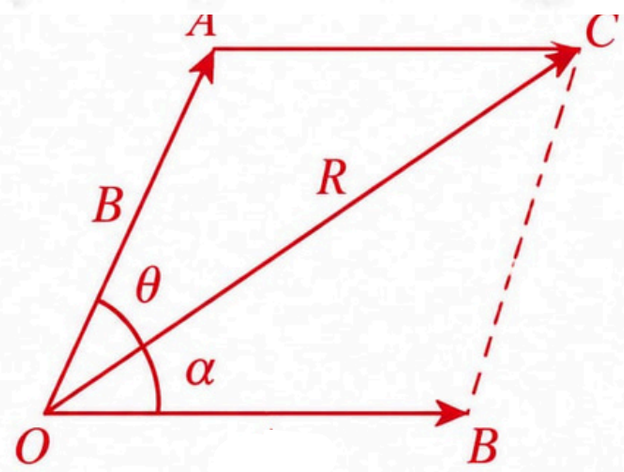
17. **Using parallelogram law of vector addition, find the magnitude and direction of the resultant of two vectors A and B in terms of their magnitudes and angle between them.**

In parallelogram OACB, the diagonal OC represents the resultant R.

From the triangle OAC, by cosine rule,

$$R^2 = A^2 + B^2 + 2AB\cos\theta$$

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$



Let α be the angle between the resultant R and vector A.

From the triangle OAC,

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

18. **A man jumping out of a slow moving bus falls forward.**

(a) **This is due to _____.** **(1)**

(b) **Which Newton's law gives the above concept ?** **(1)**

(c) **State the law.** **(1)**

(a) Inertia of motion.

(b) Newton's first law of motion.

(c) Everybody continues in its state of rest or uniform motion in a straight line unless acted upon by an external unbalanced force.

19. **What are the factors on which moment of inertia of a rigid body depends ?**

*** Mass of the body.**

*** Distribution of mass about the axis of rotation.**

*** Position and orientation of the axis of rotation.**

20. (a) Name the four processes taking place in a Carnot's cycle. (2)

(b) Can a Carnot engine work if its sink and source are interchanged ? (1)

(a) Isothermal expansion
Adiabatic expansion
Isothermal compression
Adiabatic compression

(b) No

21. (a) What are sound beats ? (1)

(b) Two sitar strings A and B playing notes are slightly out of tune and produce beats of frequency 6 Hz. The tension in the string A is slightly reduced and the beat frequency is found to be 3 Hz. If the original frequency of A is 324 Hz, what is the frequency of B ? (2)

(a) Sound beats are the periodic variations in the intensity of sound produced by the superposition of two sound waves of slightly different frequencies.

(b) $|f_A - f_B| = 6$

$f_B = 324 \pm 6$

$f_B = 318$ Hz or 330 Hz

Since the beat frequency decreases, f_B must be lower.

$\therefore \quad f_B = 318$ Hz

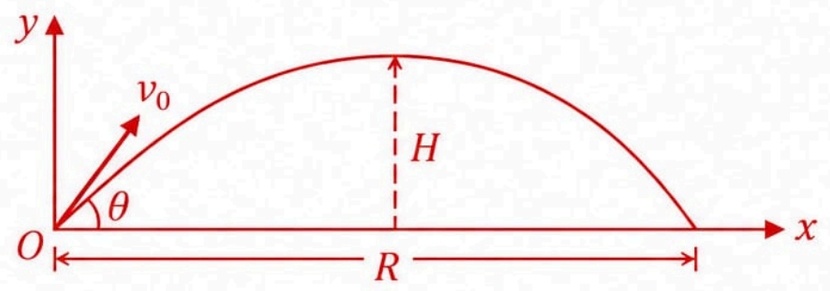
SECTION – D

Answer any 3 questions from 22 to 25. Each question carries 4 scores. **(3 × 4 = 12)**

22. A particle is projected with a velocity 'v_0' making an angle 'θ' with the horizontal. Draw the diagram and derive expressions for

(i) Time of flight

(ii) Maximum height attained



Initial vertical component: $u_y = v_0 \sin\theta$

(i) Time of flight

At the highest point, $v_y = 0$.

$$0 = v_0 \sin\theta - gt_m$$

$$t_m = \frac{v_0 \sin\theta}{g}$$

Time of flight, $T = 2t_m$

$$\therefore \quad T = \frac{2v_0 \sin\theta}{g}$$

(ii) Maximum height

Using $v_y^2 = u_y^2 + 2as$

$$0 = (v_0 \sin\theta)^2 - 2gH$$

$$\therefore \quad H = \frac{v_0^2 \sin^2\theta}{2g}$$

23. (a) State law of conservation of angular momentum. (1)

(b) A boy stands at the centre of a turntable with his arms stretched out. The table rotates at a speed of 40 rev/min. What will be the angular speed of the boy if he folds his arms, thereby reducing his moment of inertia to 2/5 of his initial value ? (3)

(a) In the absence of external torque, the total angular momentum of a system remains constant.

(b)

Initial angular speed $\omega_1 = 40$ rev/min

Initial moment of inertia I_1

Final moment of inertia $I_2 = \frac{2}{5} I_1$

Using $I_1\omega_1 = I_2\omega_2$

$$\omega_2 = \frac{I_1}{I_2}\omega_1 = \frac{I_1}{\frac{2}{5}I_1} \times 40$$

$$\omega_2 = \frac{5}{2} \times 40 = 100 \text{ rev/min}$$

24. (a) **Derive an expression for time period of oscillation of a simple pendulum assuming that the pendulum is in simple harmonic motion.** **(3)**

(b) **If this is kept on an elevator descending with an acceleration of 'g/3', find new period of oscillation.** **(1)**

(a) For small angular displacement θ, $\sin\theta \approx \theta$.

Restoring force, $F = -mg\sin\theta \approx -mg\theta$

Hence, $a = -\frac{g}{l}\theta$ (SHM)

Time period of a simple pendulum,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

(b) In an elevator descending with acceleration $\frac{g}{3}$,

effective acceleration due to gravity $g' = g - \frac{g}{3} = \frac{2g}{3}$

New time period,

$$T' = 2\pi\sqrt{\frac{l}{g'}} = 2\pi\sqrt{\frac{l}{2g/3}} = 2\pi\sqrt{\frac{3l}{2g}}$$

25. (a) **Explain the anomalous behaviour of water at 0 °C and its environmental significance.** **(2)**

(b) **Calculate the amount of heat energy required to convert 10 kg of water at 100 °C. Given latent heat L = 22.6×10^6 J/kg.** **(2)**

(a) Water shows anomalous behaviour at 0°C. At 0°C, water has maximum density at 4°C. On cooling below 4°C, it expands and its density decreases. Hence, ice is less dense than water and floats on it.

- **Environmental significance: Lakes and rivers freeze only at the surface, allowing aquatic life to survive in winter.**

(b)

Heat required to convert 10 kg of water at 100 °C to steam at 100 °C,

$$Q = mL$$
$$= 10 \times 22.6 \times 10^6$$
$$= 2.26 \times 10^8 \text{ J}$$

SECTION – E

Answer any 3 questions from 26 to 29. Each question carries 5 scores. **(3 × 5 = 15)**

26. For a particle to leave from earth's gravitational field it should be projected with a minimum velocity.

(a) Name the velocity. **(1)**

(b) Obtain an expression for it. **(2)**

(c) An elephant and an ant are to be projected into space. Whether the velocities required for doing so is same or different ? Justify. **(2)**

(a) Escape velocity

(b)

$$\frac{1}{2} m v_e^2 = \frac{GMm}{R}$$

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

(c) The velocity is the same because escape velocity is independent of the mass of the object.

27. Banking of roads helps to increase the limit of maximum safe speed of a vehicle at a curve.

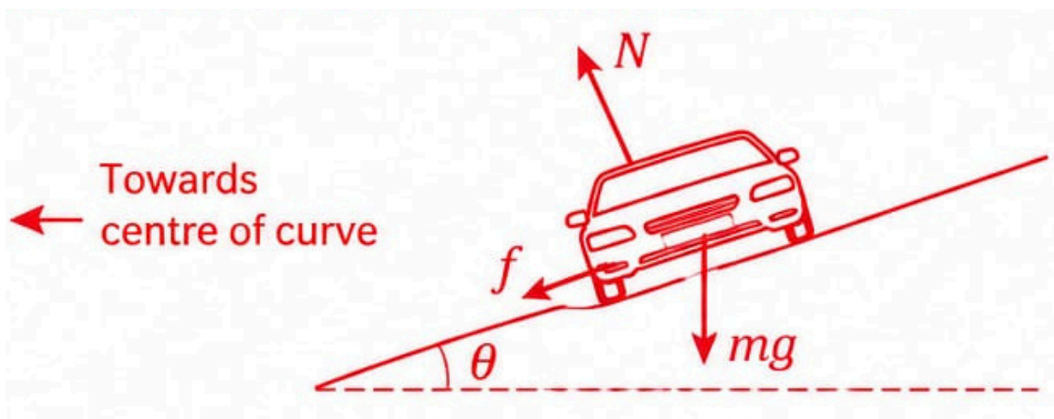
(a) What is banking of roads ? (1)

(b) Draw a schematic diagram of a vehicle on a banked road and mark various forces. (2)

(c) Obtain an expression for maximum safe speed of a vehicle on a banked road with frictional forces. (2)

(a) Banking of roads is the raising of the outer edge of a curved road above the inner edge.

(b)



(c))

Vertical direction : $N\cos\theta = mg + f\sin\theta$

Horizontal direction (towards the centre) : $N\sin\theta + f\cos\theta = \frac{mv^2}{r}$

At limiting speed, $f = \mu N$.

Substituting in (1) and (2), $N(\cos\theta - \mu\sin\theta) = mg$

$N(\sin\theta + \mu\cos\theta) = \frac{mv^2}{r}$

Dividing (4) by (3),

$$v_{max} = \sqrt{\frac{rg(\sin\theta + \mu\cos\theta)}{(\cos\theta - \mu\sin\theta)}}$$

28. (a) What is mechanical energy ? (1)

(b) Prove that total mechanical energy of a freely falling body is conserved at every point of its fall. (3)

(c) What is the nature of force so that mechanical energy is conserved ? (1)

(a) The sum of kinetic energy and potential energy of a body is called its mechanical energy.

(b) Let a body of mass m be dropped from a height h. After falling through a height x, its velocity is v.

Using the equation of motion, $v^2 = 2gx$

Kinetic energy at that point, $K = \frac{1}{2}mv^2 = \frac{1}{2}m(2gx) = mgx$

Potential energy at that point, $U = mg(h - x)$

Total mechanical energy, $E = K + U = mgx + mg(h - x) = mgh$

(c) Mechanical energy is conserved when only conservative forces act (e.g., gravitational force).

29. (a) State and prove Bernoulli's theorem in fluid flow. (3)

(b) Explain the mechanism of lift of an aerofoil using this theorem. (2)

Statement : For the steady flow of an ideal (non-viscous and incompressible) fluid, the sum of the pressure energy per unit volume, kinetic energy per unit volume and potential energy per unit volume is constant along a streamline.

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Proof : Consider a fluid element of volume ΔV moving from point 1 to point 2 along a streamline.

Work done by pressure forces on the element,

$$W = (P_1 - P_2)\,\Delta V$$

Increase in kinetic energy of the element, $\Delta K = \frac{1}{2}\rho \Delta V\,(v_2^2 - v_1^2)$

Increase in potential energy of the element, $\Delta U = \rho g \Delta V\,(h_2 - h_1)$

According to the work-energy theorem,

$$(P_1 - P_2)\,\Delta V = \frac{1}{2}\rho \Delta V\,(v_2^2 - v_1^2) + \rho g \Delta V\,(h_2 - h_1)$$

Dividing by ΔV and rearranging, $P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 = \text{constant}$

Hence $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$ (Bernoulli's theorem).

(b) Air flows faster over the curved upper surface of an aerofoil than below it. According to Bernoulli's theorem, pressure is lower above and higher below the aerofoil. This pressure difference produces an upward lift force.